NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT – 1

STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

AIM

To study computer forensics and the different tools used for forensic investigation.

PROCEDURE

1. Study the concept and importance of digital forensics in investigating cybercrimes.
2. Understand how digital evidence is collected from computer systems.
3. Study the different types of digital evidence, such as log files, registry changes, malware, and deleted files.
4. Study commercial forensic tools such as EnCase Forensic, Forensic Toolkit (FTK), and ProDiscover.
5. Examine open-source forensic tools such as The Sleuth Kit (TSK), Helix, and Knoppix.
6. Study **FTK Imager**, which is used for forensic disk imaging and file recovery.
7. Examine **Log Parser Lizard GUI**, which is used to analyze log files.
8. Study **Active@ Partition Recovery**, which is used to recover deleted partitions.

9. Examine **Autopsy**, a digital forensics platform used to investigate computer systems and recover digital evidence.
10. Understand how forensic tools help collect, examine, and analyze digital evidence during an investigation.

OBSERVATION

Digital forensics involves identifying, collecting, examining, and analyzing digital evidence. Various commercial and open-source tools provide features for disk imaging, log analysis, file recovery, and computer investigations.

RESULT

Thus, computer forensics and the different tools used for forensic investigation were studied successfully.

NAME: SREE DHARSHINI M

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EXPERIMENT – 2

LIVE FORENSICS CASE INVESTIGATION USING AUTOPSY

AIM

To investigate a live forensic case using Autopsy and analyze digital evidence from a computer system or storage device.

PROCEDURE

1. Open Autopsy and click **New Case**. Enter the case name and base directory.
2. Enter the case number and examiner details, then click **Finish**.
3. Select the appropriate data source and browse to the evidence file or device.
4. Configure the required **Ingest Modules** and click **Finish** to start processing.
5. Examine the forensic investigation results after processing is complete.
6. View device details, EXIF metadata, installed programs, operating system information, and user accounts.
7. Analyze recent documents, web bookmarks, cookies, downloads, browsing history, search history, and email addresses.
8. Review the extracted artifacts and document the findings.

OBSERVATION

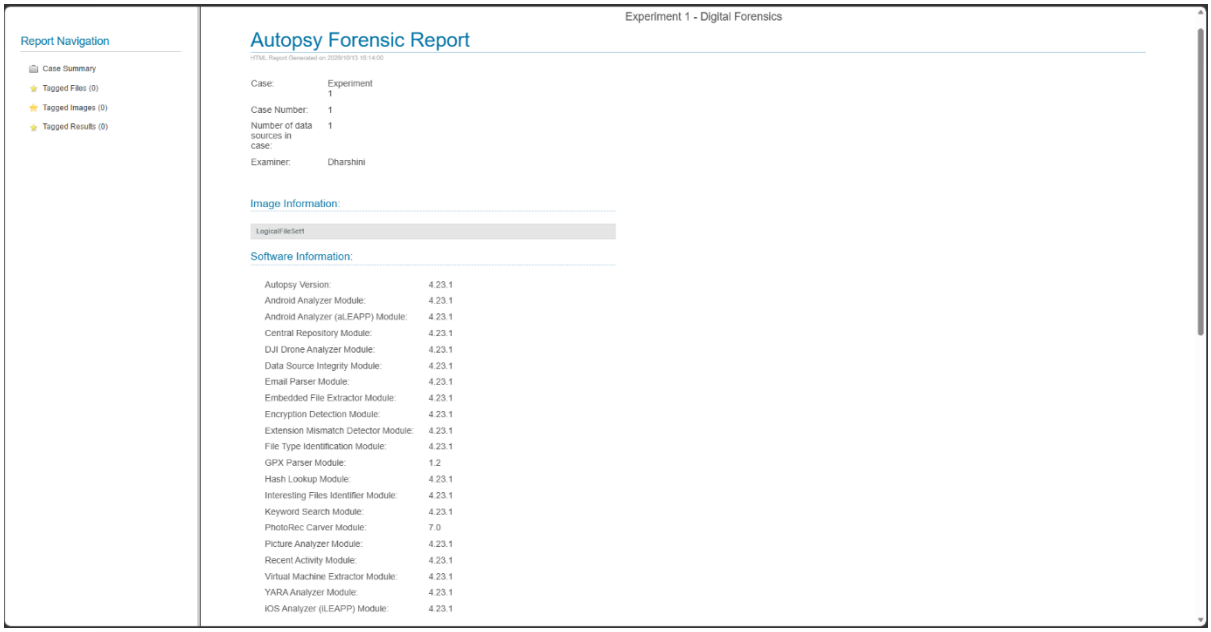


Figure 1.1 Autopsy Forensic Report

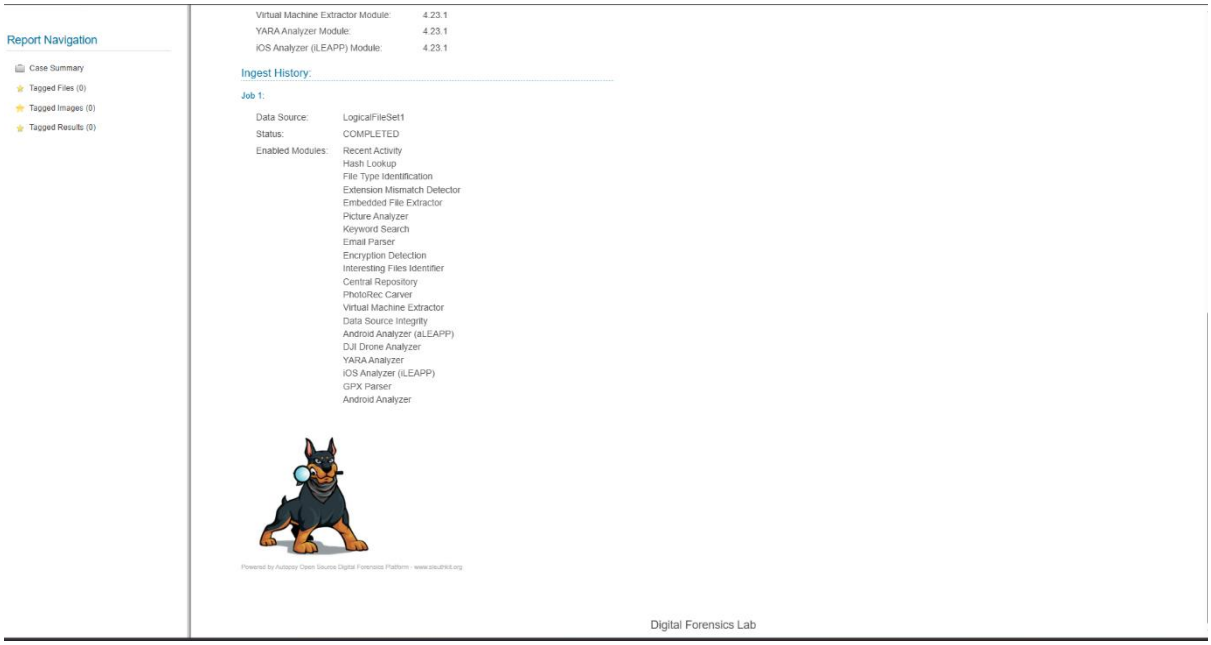


Figure 1.2 Autopsy Forensic Report

RESULT

Thus, Autopsy was used to create a forensic case, process the selected evidence source, and examine digital artifacts for forensic investigation.

NAME: SREE DHARSHINI M

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EXPERIMENT 3

RECOVER DELETED FILES USING FORENSICS TOOLS

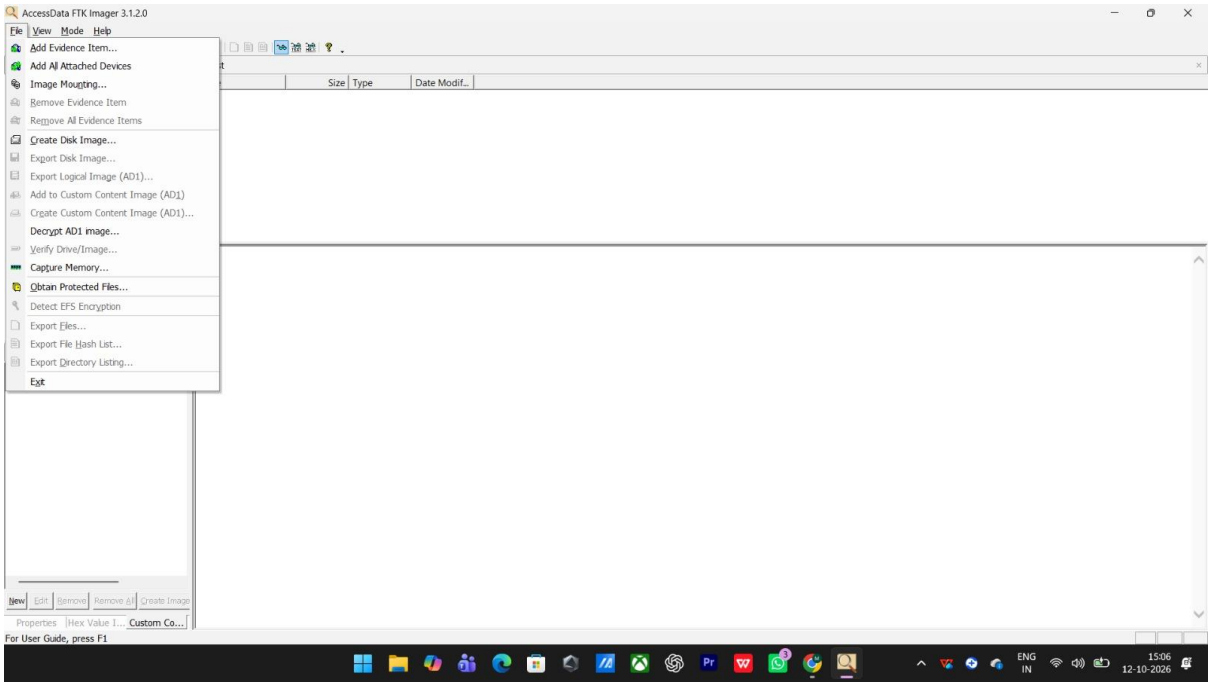
AIM

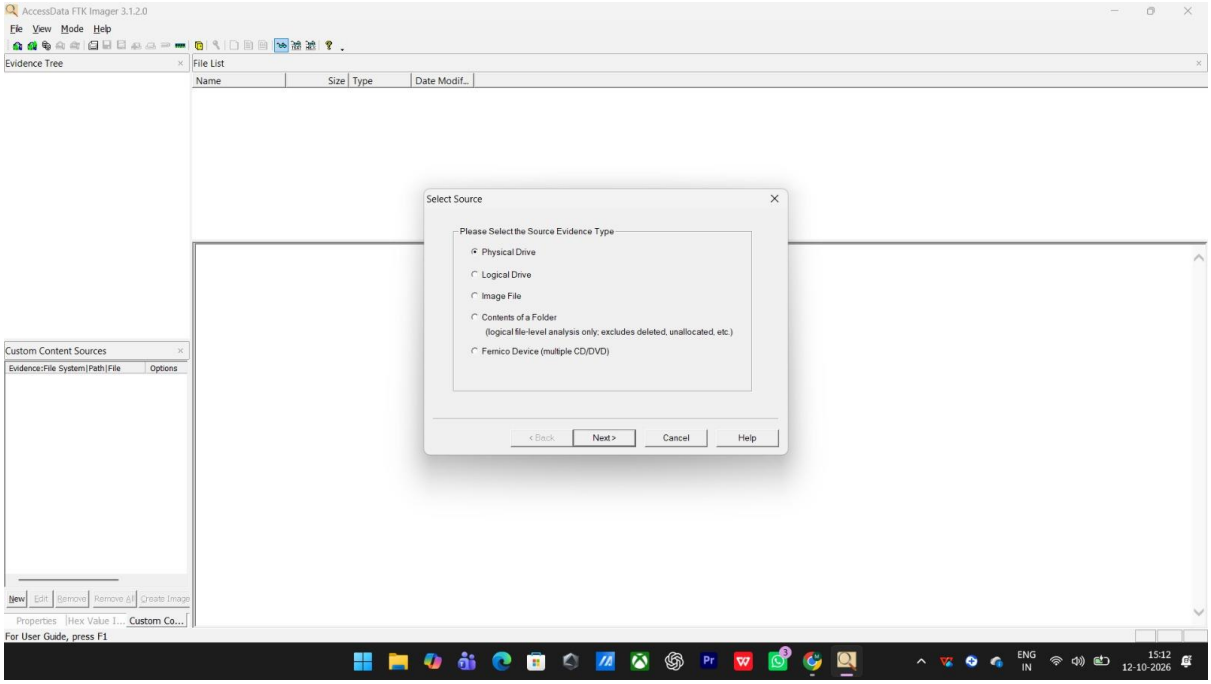
To recover deleted files from a storage device using forensic tools and examine the recovered data.

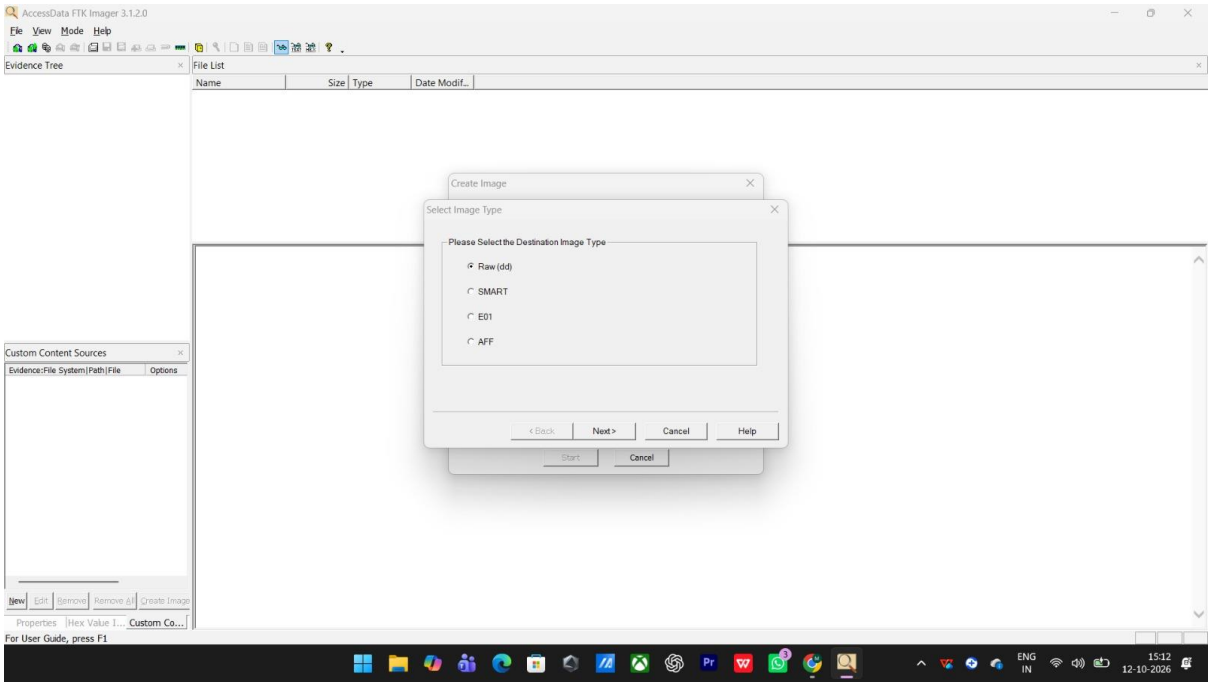
PROCEDURE

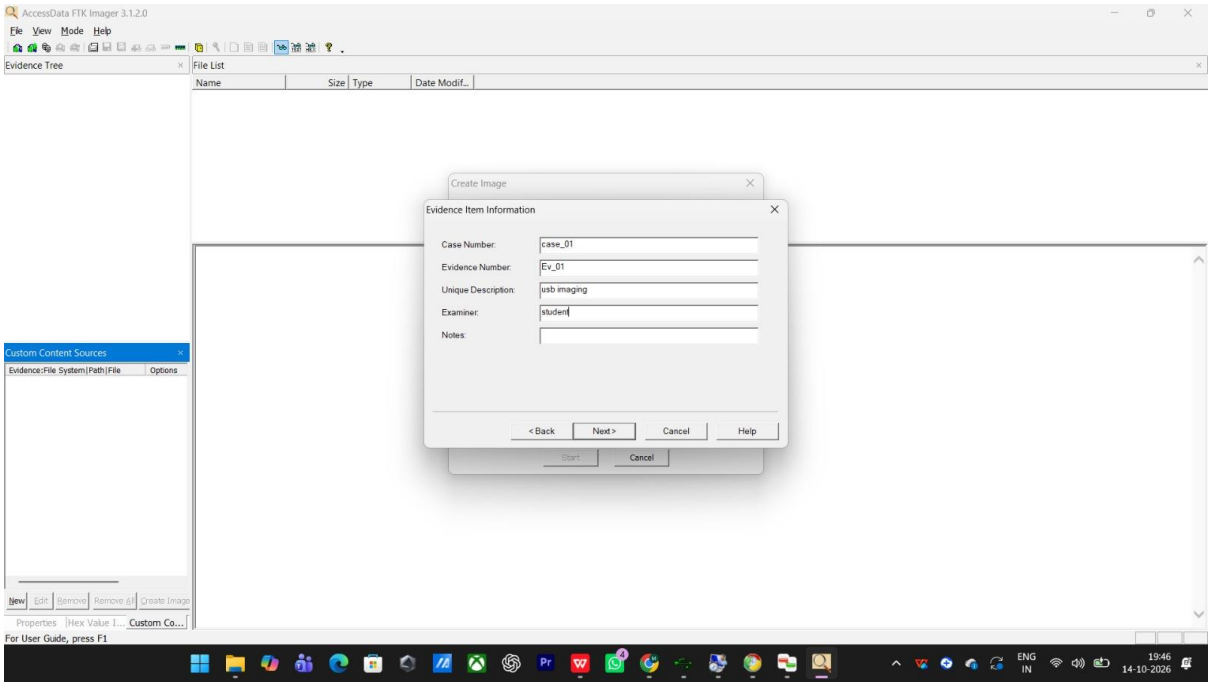
1. Select a storage device or disk image for the file recovery process.
2. Open a suitable forensic recovery tool.
3. Select the drive or storage location from which the deleted files need to be recovered.
4. Scan the selected storage device for deleted files and folders.
5. Wait for the scanning process to complete.
6. Examine the list of deleted files identified by the software.
7. Select the required files that are available for recovery.
8. Choose a separate destination folder to save the recovered files.
9. Recover the selected files and verify whether they can be opened successfully.

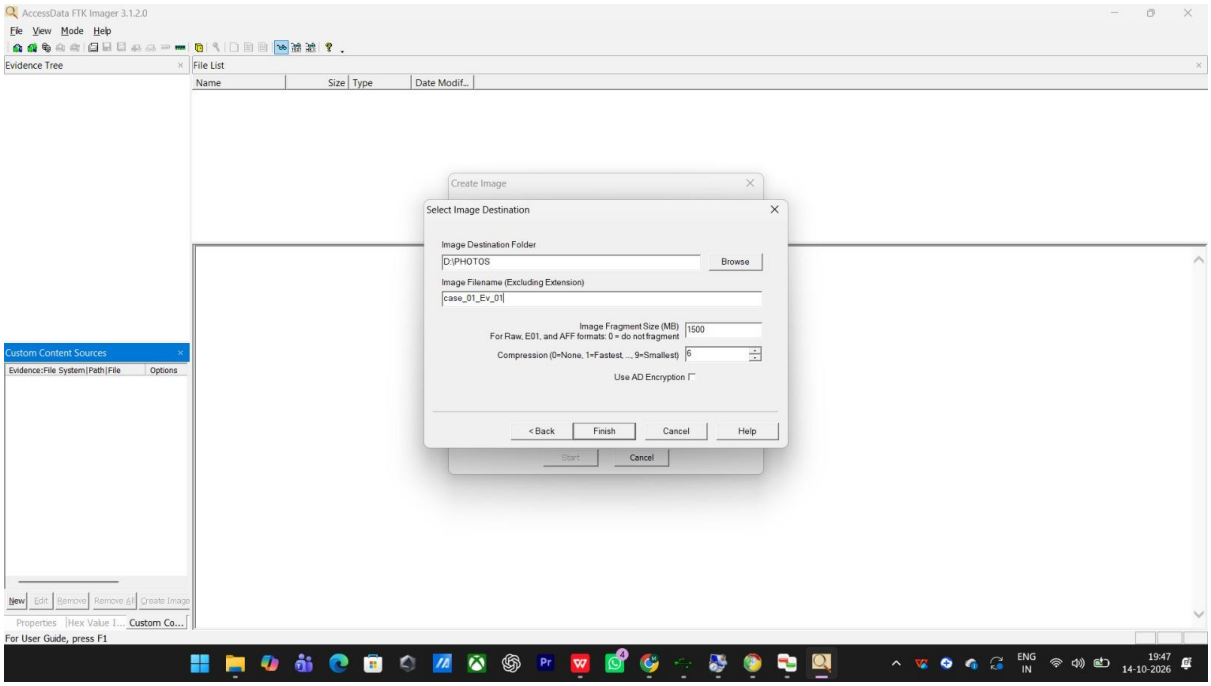
OBSERVATION

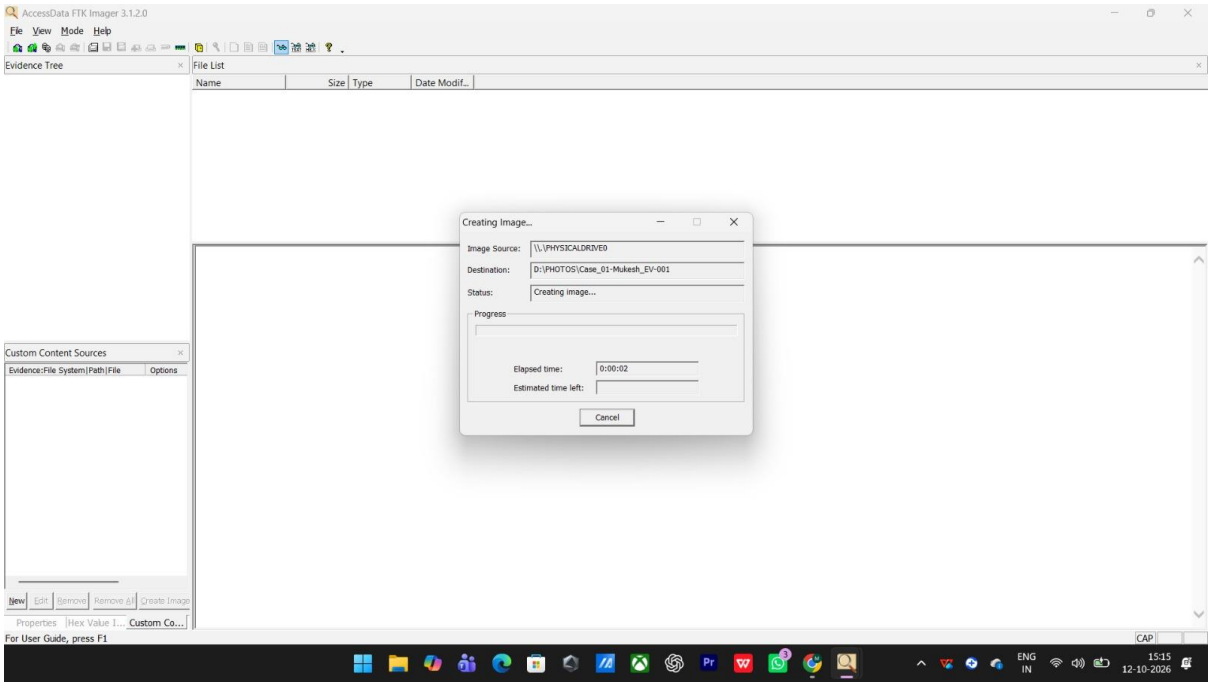

AccessData FTK Imager 3.1.2.0
File View Mode Help
Add Evidence Item...
Add All Attached Devices
Image Mounting...
Remove Evidence Item
Remove All Evidence Items
Create Disk Image...
Export Disk Image...
Export Logical Image (AD1)...
Add to Custom Content Image (AD1)
Create Custom Content Image (AD1)...
Decrypt AD1 image...
Verify Drive/Image...
Capture Memory...
Obtain Protected Files...
Detect EFS Encryption
Export Files...
Export File Hash List...
Export Directory Listing...
Exit
Size
Type
Date Modif...
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value I...
Custom Co...
For User Guide, press F1
ENG
IN
15:06
12-10-2026

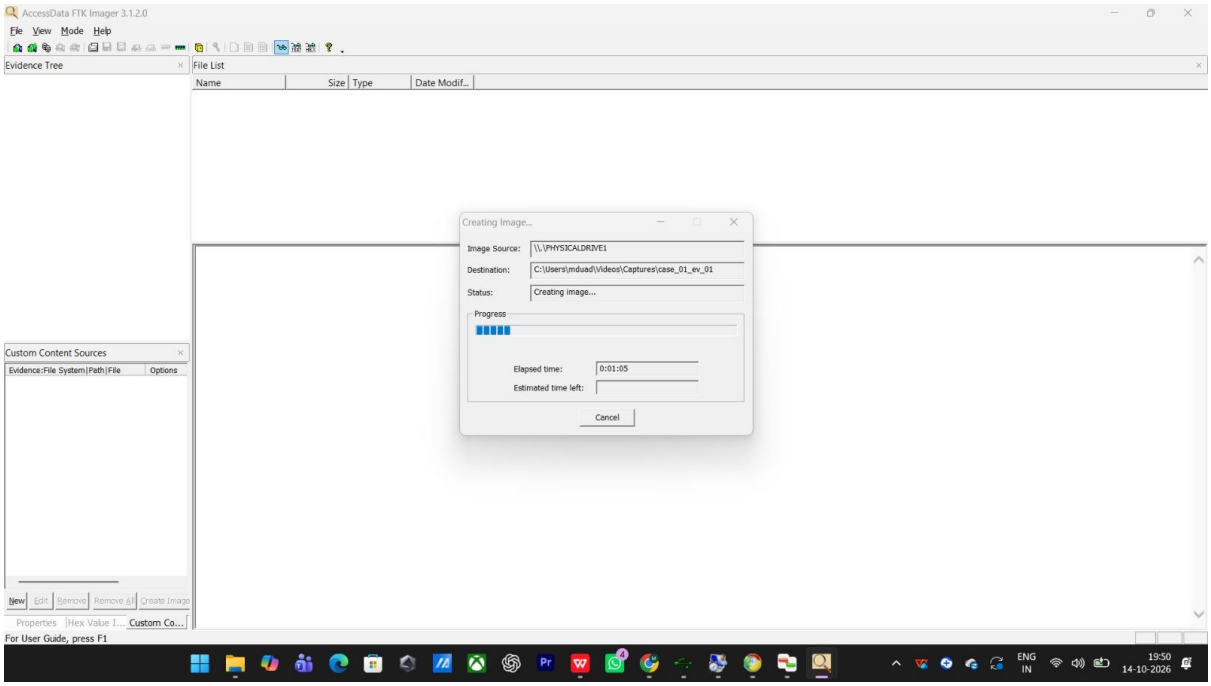

AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name
Size
Type
Date Modif...
Select Source
Please Select the Source Evidence Type
Physical Drive
Logical Drive
Image File
Contents of a Folder
(logical file-level analysis only; excludes deleted, unallocated, etc.)
Fernico Device (multiple CD/DVD)
< Back
Next >
Cancel
Help
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value I...
Custom Co...
For User Guide, press F1
ENG
IN
15:12
12-10-2026

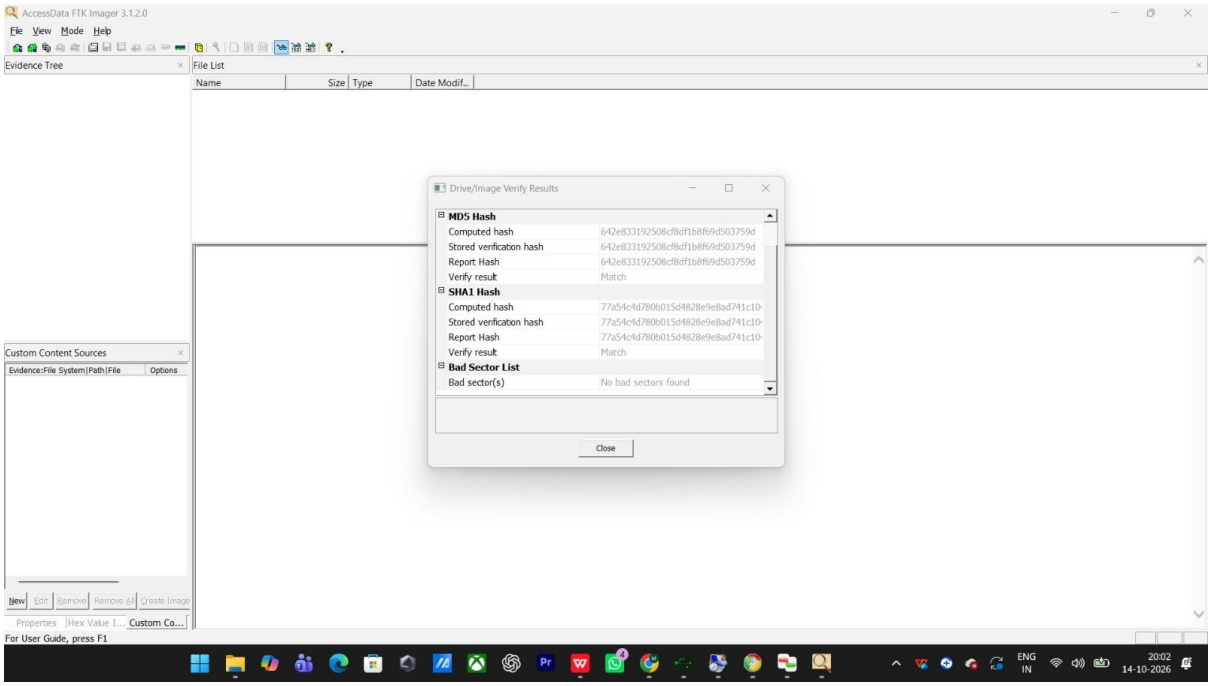

AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name Size Type Date Modif...
Create Image
Select Image Type
Please Select the Destination Image Type
Raw (dd)
SMART
E01
AFF
< Back Next > Cancel Help
Start Cancel
Custom Content Sources
Evidence:File System|Path|File Options
New Edit Remove Remove All Create Image
Properties Hex Value I... Custom Co...
For User Guide, press F1
ENG IN
15:12
12-10-2026


AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name Size Type Date Modif...
Create Image
Evidence Item Information
Case Number: case_01
Evidence Number: Ev_01
Unique Description: usb imaging
Examiner: student
Notes:
< Back Next > Cancel Help
Start Cancel
Custom Content Sources
Evidence:File System|Path|File Options
New Edit Remove Remove All Create Image
Properties Hex Value I... Custom Co...
For User Guide, press F1
ENG IN
19:46
14-10-2026


AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name
Size
Type
Date Modif...
Create Image
Select Image Destination
Image Destination Folder
D:\PHOTOS
Browse
Image Filename (Excluding Extension)
case_01_Ev_01
Image Fragment Size (MB)
For Raw, E01, and AFF formats: 0 = do not fragment
1500
Compression (0=None, 1=Fastest, ... 9=Smallest)
6
Use AD Encryption
< Back
Finish
Cancel
Help
Start
Cancel
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value I...
Custom Co...
For User Guide, press F1
ENG
IN
19:47
14-10-2026


AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name
Size
Type
Date Modif...
Creating Image...
Image Source:
\\.\PHYSICALDRIVE0
Destination:
D:\PHOTOS\Case_01-Mukesh_EV-001
Status:
Creating image...
Progress
Elapsed time:
0:00:02
Estimated time left:
Cancel
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value I...
Custom Co...
For User Guide, press F1
CAP
ENG
IN
15:15
12-10-2026


AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name
Size
Type
Date Modif...
Creating Image...
Image Source: \\.\PHYSICALDRIVE1
Destination: C:\Users\mduad\Videos\Captures\case_01_ev_01
Status: Creating image...
Progress
Elapsed time: 0:01:05
Estimated time left:
Cancel
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value I...
Custom Co...
For User Guide, press F1
ENG
IN
19:50
14-10-2026


AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name
Size
Type
Date Modif...
Drive/Image Verify Results
MD5 Hash
Computed hash 642e833192508cf8df1b8f69d503759d
Stored verification hash 642e833192508cf8df1b8f69d503759d
Report Hash 642e833192508cf8df1b8f69d503759d
Verify result Match
SHA1 Hash
Computed hash 77a54c4d780b015d4828e9e8ad741c10
Stored verification hash 77a54c4d780b015d4828e9e8ad741c10
Report Hash 77a54c4d780b015d4828e9e8ad741c10
Verify result Match
Bad Sector List
Bad sector(s) No bad sectors found
Close
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value I...
Custom Co...
For User Guide, press F1
ENG
IN
20:02
14-10-2026

RESULT

Deleted files were searched for and recovered using forensic recovery tools, and the recovered data was examined successfully.

NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT – 4

FIND LAST CONNECTED USB ON YOUR SYSTEM (USB FORENSICS)

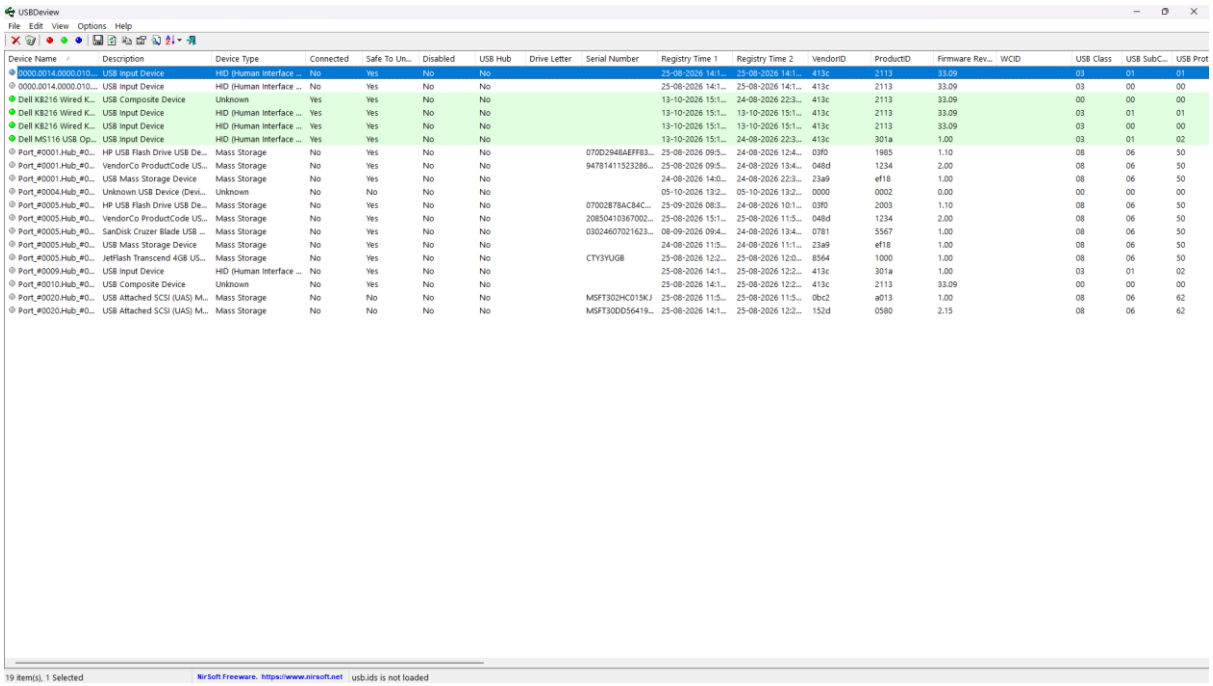
AIM

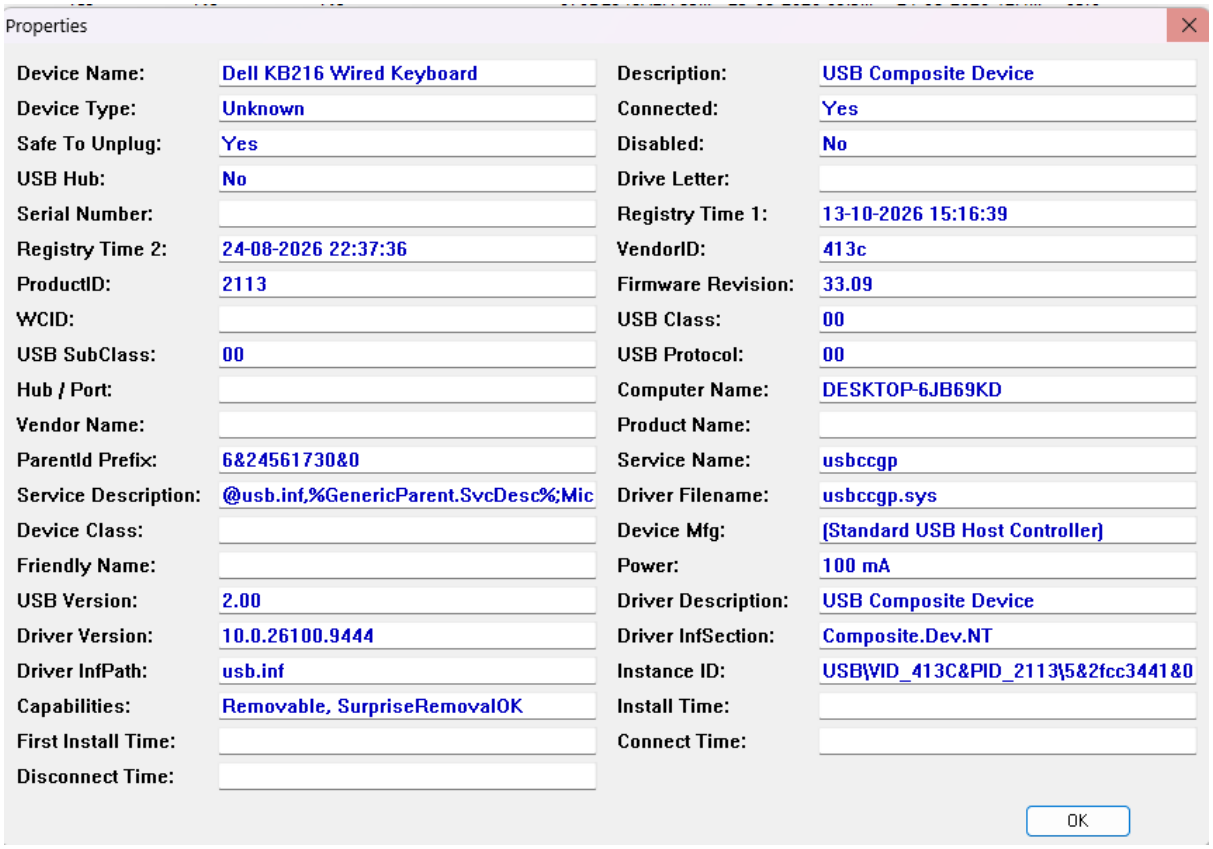
To identify the last connected USB device on a system using USB forensics.

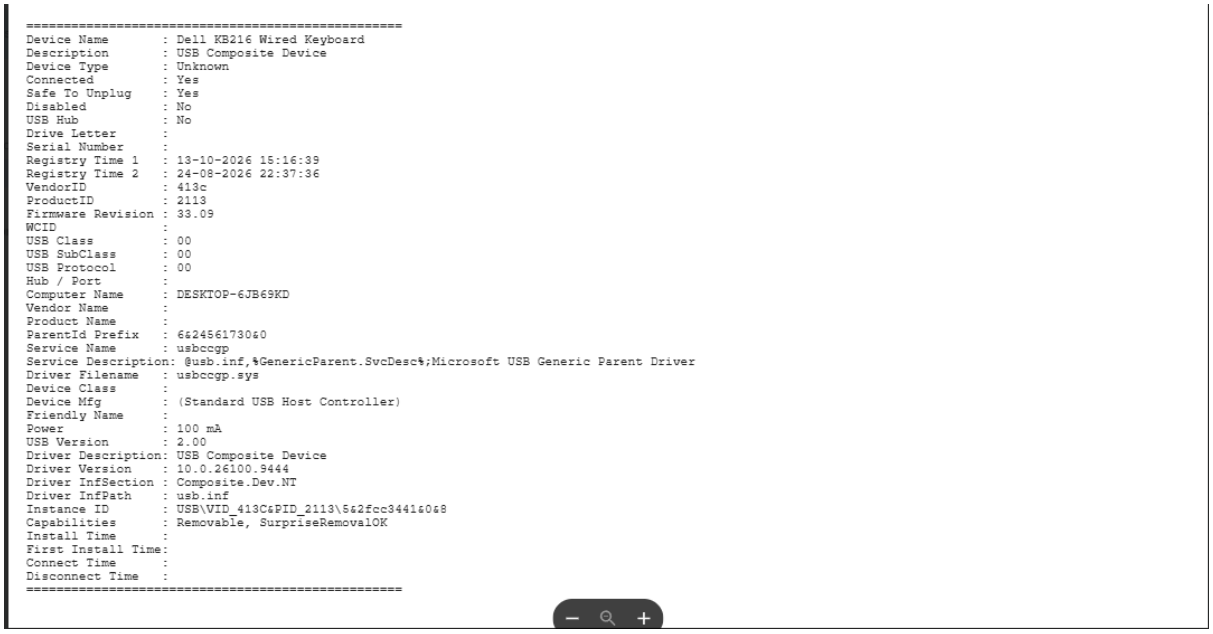
PROCEDURE

1. Download and run the **USBDeview** utility on the Windows system.
2. Open USBDeview to view the list of USB devices.
3. Identify the USB devices currently connected to the computer.
4. Examine the list of previously used USB devices.
5. Check the device name and description of each USB device.
6. Observe the device type and serial number, if available.
7. Check the date and time when each USB device was added to the system.
8. Examine the Vendor ID and Product ID of the USB devices.
9. Compare the available device details and timestamps to identify the most recently connected USB device.

OBSERVATION







```
==================================================
Device Name       : Dell KB216 Wired Keyboard
Description       : USB Composite Device
Device Type       : Unknown
Connected         : Yes
Safe To Unplug    : Yes
Disabled          : No
USB Hub           : No
Drive Letter      :
Serial Number     :
Registry Time 1   : 13-10-2026 15:16:39
Registry Time 2   : 24-08-2026 22:37:36
VendorID          : 413c
ProductID         : 2113
Firmware Revision : 33.09
WCID              :
USB Class         : 00
USB SubClass      : 00
USB Protocol      : 00
Hub / Port        :
Computer Name     : DESKTOP-6JB69KD
Vendor Name       :
Product Name      :
ParentId Prefix   : 6&24561730&0
Service Name      : usbccgp
Service Description: @usb.inf,%GenericParent.SvcDesc%;Microsoft USB Generic Parent Driver
Driver Filename   : usbccgp.sys
Device Class      :
Device Mfg        : (Standard USB Host Controller)
Friendly Name     :
Power             : 100 mA
USB Version       : 2.00
Driver Description: USB Composite Device
Driver Version    : 10.0.26100.9444
Driver InfSection : Composite.Dev.NT
Driver InfPath    : usb.inf
Instance ID       : USB\VID_413C&PID_2113\5&2fcc3441&0&8
Capabilities      : Removable, SurpriseRemovalOK
Install Time      :
First Install Time:
Connect Time      :
Disconnect Time   :
==================================================
```

RESULT

Thus, USBDeview was used to examine USB device details and identify the most recently connected USB device on the system.

NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT – 5

VIEW LAST ACTIVITY OF YOUR PC

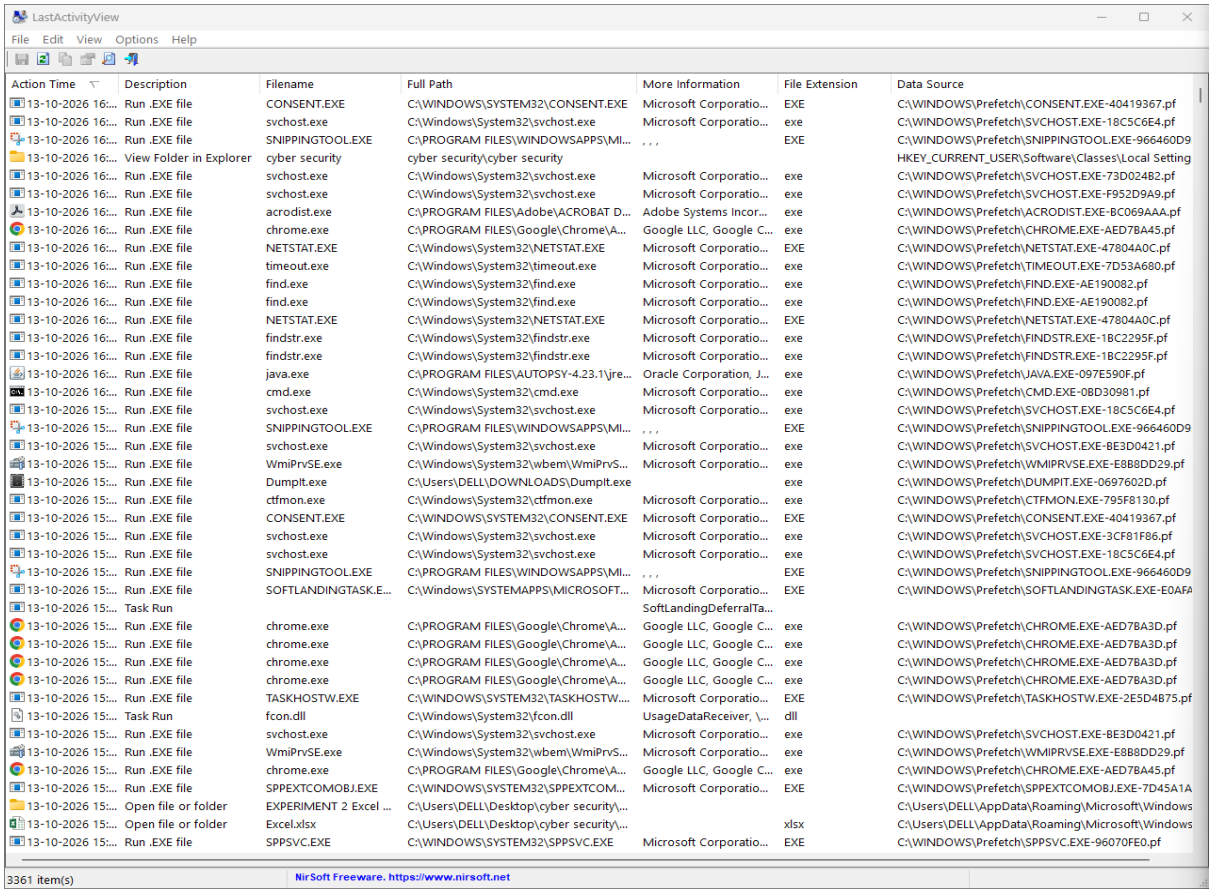
AIM

To view the last activity of a PC using the LastActivityView tool.

PROCEDURE

1. Download the **LastActivityView** tool for the Windows operating system.
2. Extract the downloaded files, if required.
3. Open the LastActivityView application.
4. Allow the tool to collect activity information from the system.
5. View the list of user actions and events displayed in the application.
6. Examine the activity descriptions to understand the recorded events.
7. Check the available date and time details for the recorded activities.
8. Analyze the displayed information to identify the recent activities performed on the PC.

OBSERVATION




```
==================================================
Action Time       : 13-10-2026 16:01:13
Description       : Run .EXE file
Filename          : svchost.exe
Full Path         : C:\Windows\System32\svchost.exe
More Information  : Microsoft Corporation, Microsoft® Windows® Operating System, Host Process for Windows Services, 10.0.26100.1 (WinBuild.160101.0800)
File Extension    : exe
Data Source       : C:\WINDOWS\Prefetch\SVCHOST.EXE-73D024B2.pf
==================================================

==================================================
Action Time       : 13-10-2026 16:01:13
Description       : Run .EXE file
Filename          : svchost.exe
Full Path         : C:\Windows\System32\svchost.exe
More Information  : Microsoft Corporation, Microsoft® Windows® Operating System, Host Process for Windows Services, 10.0.26100.1 (WinBuild.160101.0800)
File Extension    : exe
Data Source       : C:\WINDOWS\Prefetch\SVCHOST.EXE-F952D9A9.pf
==================================================

==================================================
Action Time       : 13-10-2026 16:01:09
Description       : Run .EXE file
Filename          : acrodist.exe
Full Path         : C:\PROGRAM FILES\Adobe\ACROBAT DC\Acrobat\acrodist.exe
More Information  : Adobe Systems Incorporated., Acrobat  Distiller for Windows, Acrobat  Distiller, 24.5.20320.0
File Extension    : exe
Data Source       : C:\WINDOWS\Prefetch\ACRODIST.EXE-BC069AAA.pf
==================================================

==================================================
Action Time       : 13-10-2026 16:00:55
Description       : Run .EXE file
Filename          : chrome.exe
Full Path         : C:\PROGRAM FILES\Google\Chrome\APPLICATION\chrome.exe
More Information  : Google LLC, Google Chrome, Google Chrome, 155.0.8059.40
File Extension    : exe
Data Source       : C:\WINDOWS\Prefetch\CHROME.EXE-AED7BA45.pf
==================================================

==================================================
Action Time       : 13-10-2026 16:00:07
Description       : Run .EXE file
Filename          : NETSTAT.EXE
Full Path         : C:\Windows\System32\NETSTAT.EXE
More Information  : Microsoft Corporation, Microsoft® Windows® Operating System, TCP/IP Netstat Command, 10.0.26100.8521 (WinBuild.160101.0800)
File Extension    : EXE
Data Source       : C:\WINDOWS\Prefetch\NETSTAT.EXE-47804A0C.pf
==================================================

==================================================
Action Time       : 13-10-2026 16:00:07
Description       : Run .EXE file
Filename          : timeout.exe
```

RESULT

Thus, the LastActivityView tool was used to examine and view the recorded activities and events of the PC.

NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT – 6

EXTRACTION OF BROWSER ARTIFACTS

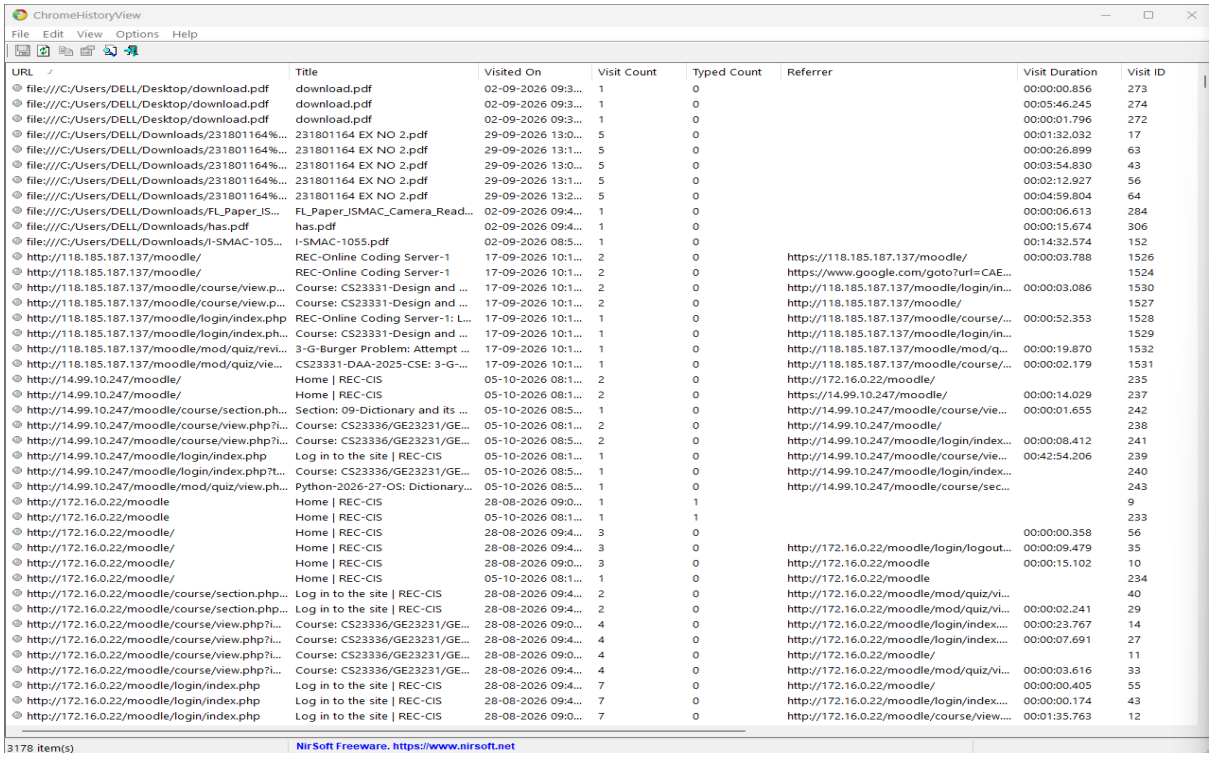
AIM

To extract and examine browser artifacts using forensic tools.

PROCEDURE

1. Use **ChromeHistoryView** to examine Google Chrome browsing history.
2. Open the tool and view the list of visited web pages.
3. Observe the URL, page title, visit date and time, and number of visits.
4. Examine the typed count, referrer, and visit ID details.
5. Use **ChromeCacheView** to examine files stored in the Chrome cache folder.
6. Observe the URL, content type, file size, and last accessed time of cached files.
7. Use **IEHistoryView** to examine the browsing history stored by Internet Explorer.
8. Use **IECacheView** to examine cached files and their details, including filename, URL, last accessed time, and file size.
9. Analyze the collected browser artifacts for forensic investigation.

OBSERVATION




```
==================================================
URL                 : file:///C:/Users/DELL/Desktop/download.pdf
Title               : download.pdf
Visited On          : 02-09-2026 09:34:47
Visit Count         : 1
Typed Count         : 0
Referrer            :
Visit Duration      : 00:00:00.856
Visit ID            : 273
Profile             : Default
URL Length          : 42
Transition Type     : Reload
Transition Qualifiers: Chain Start,Chain End
History File        : C:\Users\DELL\AppData\Local\Google\Chrome\User Data\Default\History
==================================================

==================================================
URL                 : file:///C:/Users/DELL/Desktop/download.pdf
Title               : download.pdf
Visited On          : 02-09-2026 09:34:48
Visit Count         : 1
Typed Count         : 0
Referrer            :
Visit Duration      : 00:05:46.245
Visit ID            : 274
Profile             : Default
URL Length          : 42
Transition Type     : Reload
Transition Qualifiers: Chain Start,Chain End
History File        : C:\Users\DELL\AppData\Local\Google\Chrome\User Data\Default\History
==================================================

==================================================
URL                 : file:///C:/Users/DELL/Desktop/download.pdf
Title               : download.pdf
Visited On          : 02-09-2026 09:34:45
Visit Count         : 1
Typed Count         : 0
Referrer            :
Visit Duration      : 00:00:01.796
Visit ID            : 272
Profile             : Default
URL Length          : 42
Transition Type     : Link
Transition Qualifiers: Chain Start,Chain End
History File        : C:\Users\DELL\AppData\Local\Google\Chrome\User Data\Default\History
==================================================
```

RESULT

Thus, browser artifacts were examined using forensic tools to analyze browsing history and cached data.

NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT – 7

COMPARISON OF TWO FILES FOR FORENSICS INVESTIGATION BY COMPARE IT SOFTWARE

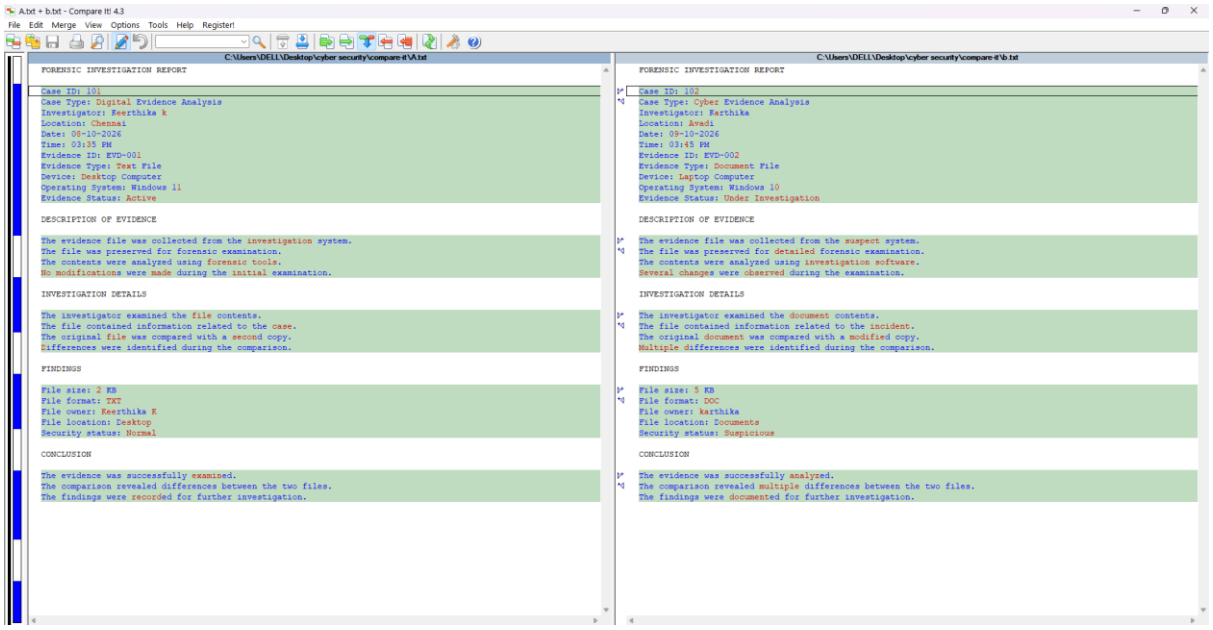
AIM

To compare two files for forensic investigation using Compare It software.

PROCEDURE

1. Download and install the **Compare It** software.
2. Open the Compare It application.
3. Select the option to choose files for comparison.
4. Browse and select the first file, then click **Open**.
5. Select the second file and click **Open**.
6. Allow the software to compare the contents of both files.
7. Observe the differences highlighted in the comparison window.
8. Examine the changes displayed side by side.
9. Generate or print a report showing the differences between the two files.

OBSERVATION




```
FORENSIC INVESTIGATION REPORT

Case ID: 101
Case Type: Digital Evidence Analysis
Investigator: Keerthika k
Location: Chennai
Date: 08-10-2026
Time: 03:35 PM
Evidence ID: EVD-001
Evidence Type: Text File
Device: Desktop Computer
Operating System: Windows 11
Evidence Status: Active

DESCRIPTION OF EVIDENCE

The evidence file was collected from the investigation system.
The file was preserved for forensic examination.
The contents were analyzed using forensic tools.
No modifications were made during the initial examination.

INVESTIGATION DETAILS

The investigator examined the file contents.
The file contained information related to the case.
The original file was compared with a second copy.
Differences were identified during the comparison.

FINDINGS

File size: 2 KB
File format: TXT
File owner: Keerthika K
File location: Desktop
Security status: Normal

CONCLUSION

The evidence was successfully examined.
The comparison revealed differences between the two files.
The findings were recorded for further investigation.
```

```
FORENSIC INVESTIGATION REPORT

Case ID: 102
Case Type: Cyber Evidence Analysis
Investigator: Karthika
Location: Avadi
Date: 09-10-2026
Time: 03:45 PM
Evidence ID: EVD-002
Evidence Type: Document File
Device: Laptop Computer
Operating System: Windows 10
Evidence Status: Under Investigation

DESCRIPTION OF EVIDENCE

The evidence file was collected from the suspect system.
The file was preserved for detailed forensic examination.
The contents were analyzed using investigation software.
Several changes were observed during the examination.

INVESTIGATION DETAILS

The investigator examined the document contents.
The file contained information related to the incident.
The original document was compared with a modified copy.
Multiple differences were identified during the comparison.

FINDINGS

File size: 5 KB
File format: DOC
File owner: karthika
File location: Documents
Security status: Suspicious

CONCLUSION

The evidence was successfully analyzed.
The comparison revealed multiple differences between the two files.
The findings were documented for further investigation.
```

RESULT

Thus, two files were compared successfully using Compare It software, and the differences were identified for forensic investigation.

NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT 8

COLLECTING EMAIL EVIDENCE IN VICTIM PC

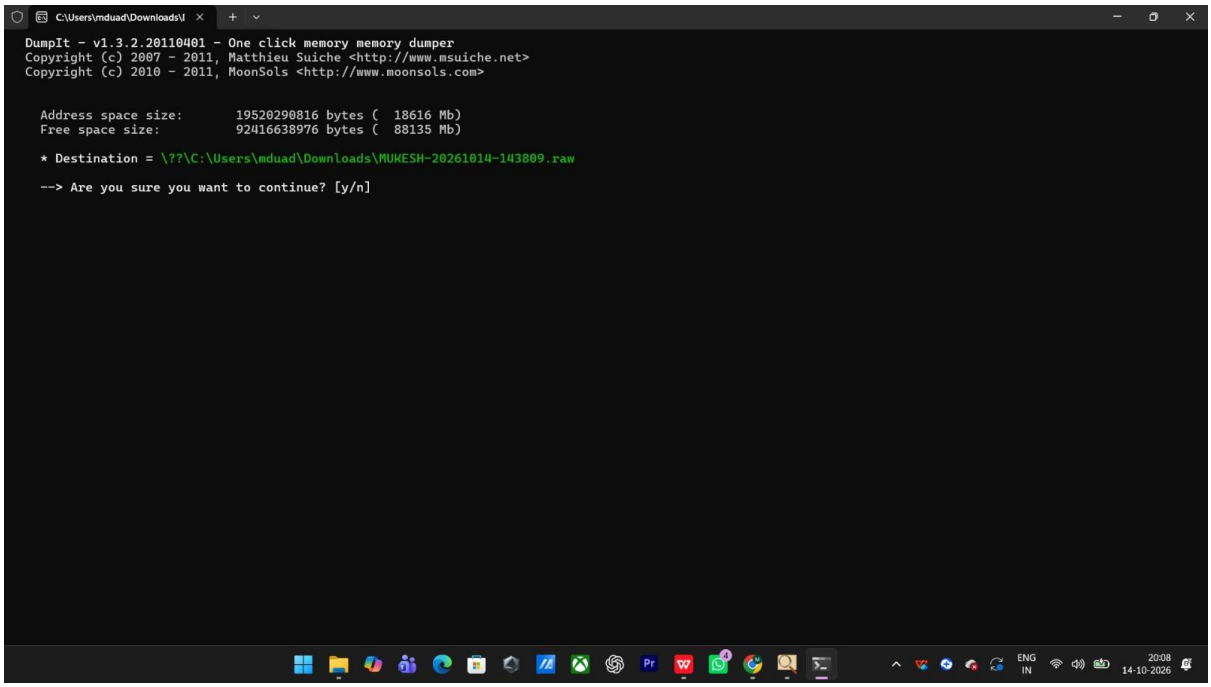
AIM

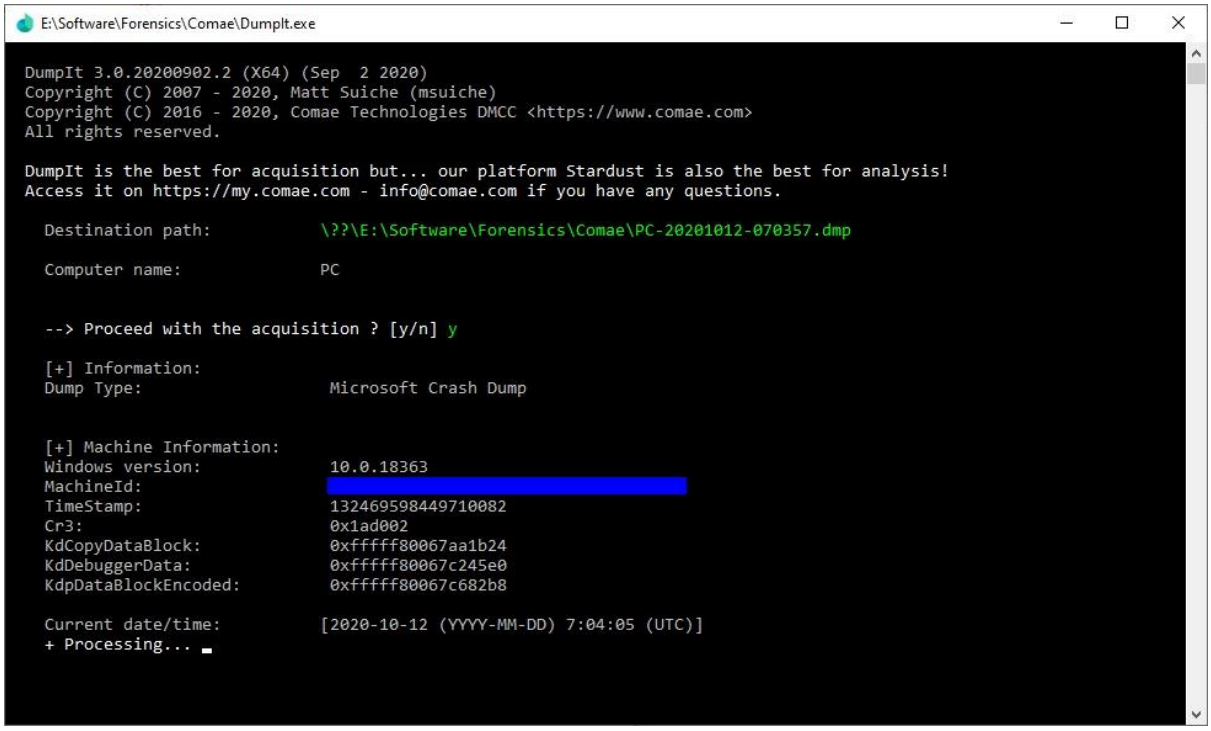
To collect and examine email evidence from a victim's computer for digital forensic investigation.

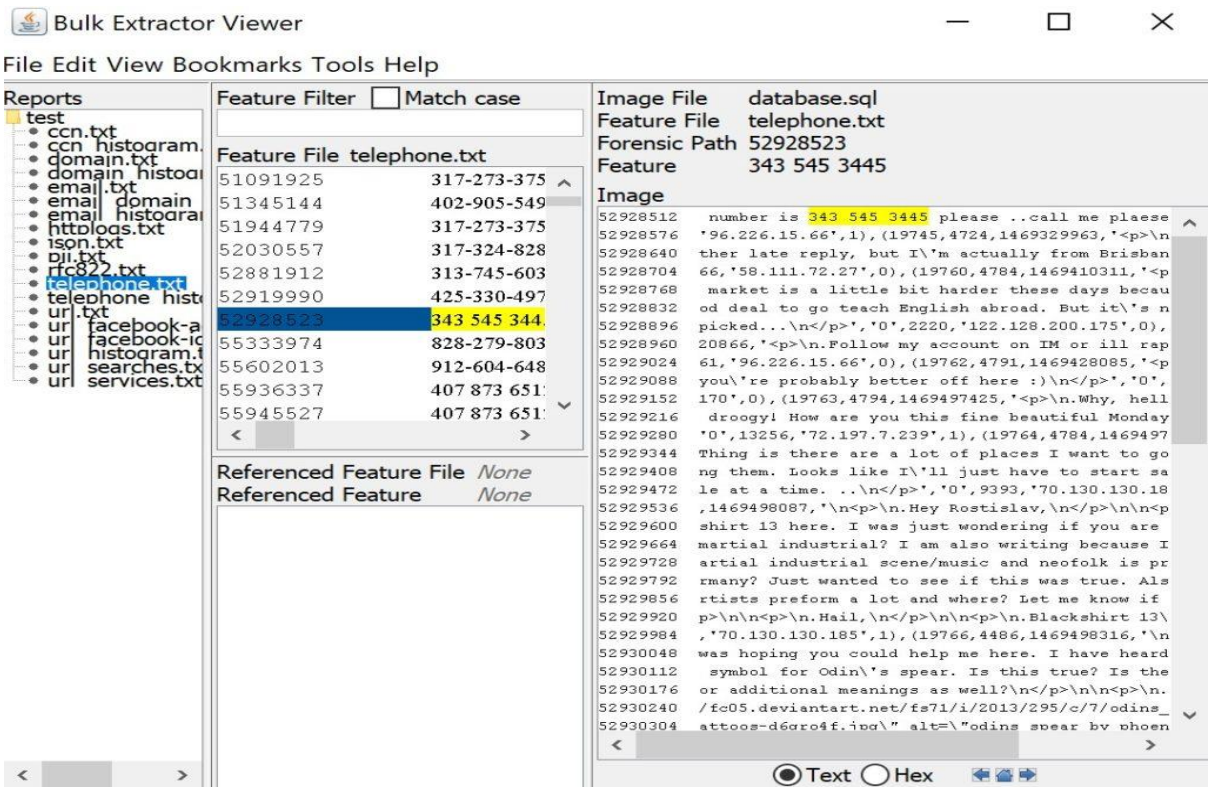
PROCEDURE

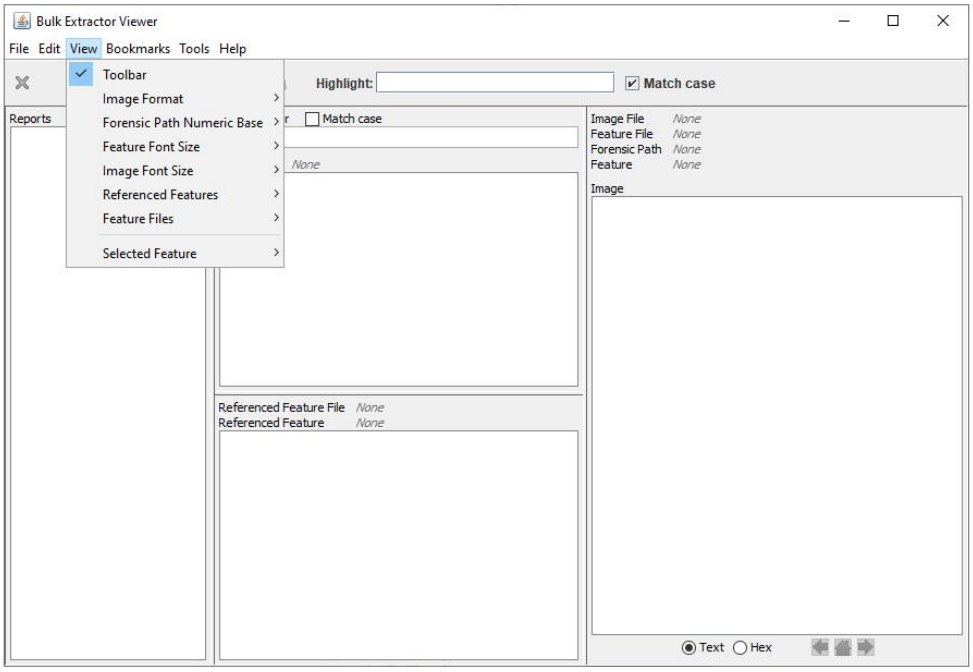
1. Identify the computer or forensic image from which email evidence needs to be collected.
2. Open a suitable digital forensic tool.
3. Locate the email storage files or folders on the system.
4. Identify the email client used on the computer, if applicable.
5. Search for stored emails and related email data.
6. Examine email details such as sender, receiver, subject, date, and time.
7. Inspect email attachments and related files, where available.
8. Preserve the collected email evidence without modifying the original data.
9. Record the findings and document relevant email evidence for further investigation.

OBSERVATION









SoftRadar.com
Software reviews & downloads

RESULT

Email evidence was collected and examined from the victim's computer for digital forensic investigation.

NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT – 9

STUDY THE STEPS FOR HIDING AND EXTRACTING ANY TEXT FILE BEHIND AN IMAGE FILE/AUDIO FILE USING COMMAND PROMPT.

AIM

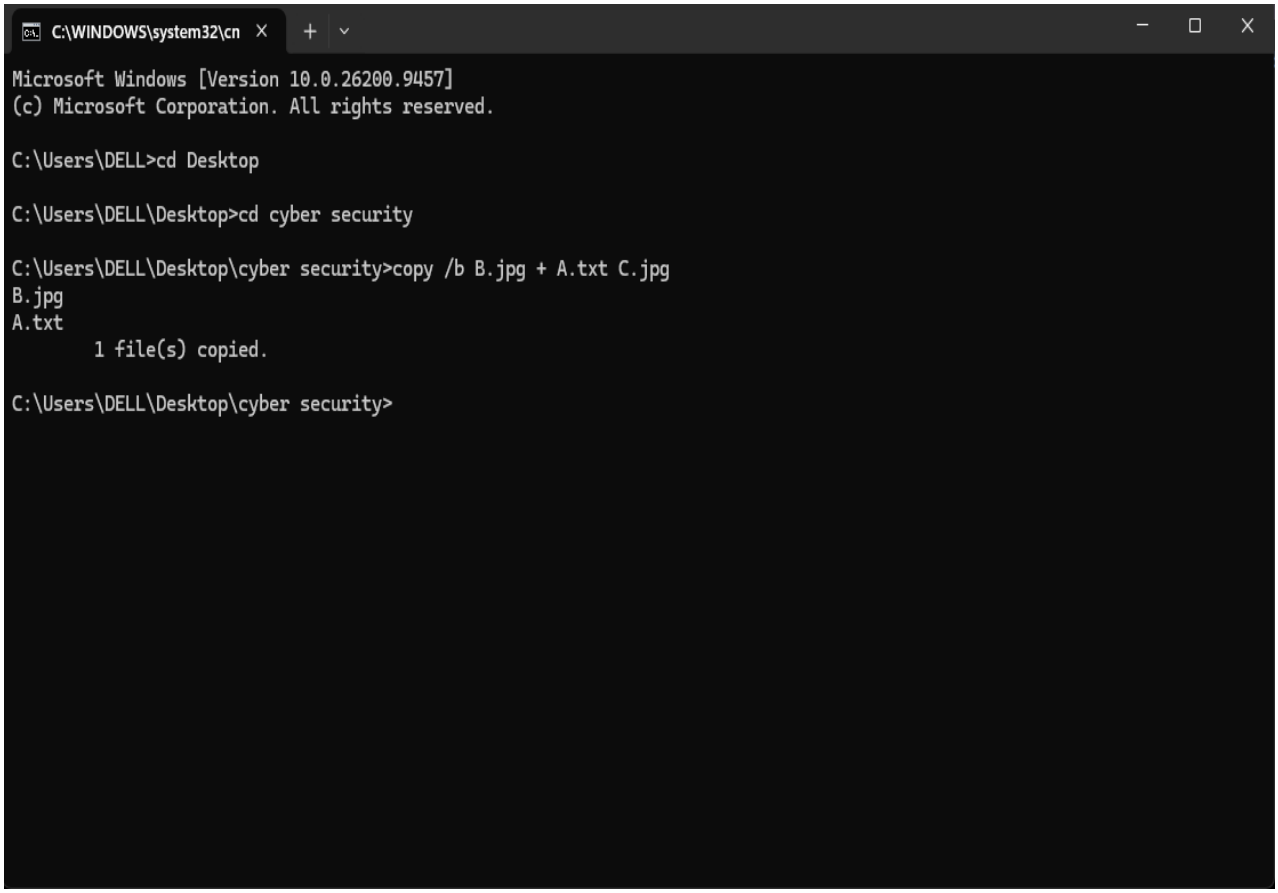
To study the steps for hiding and extracting a text file behind an image or audio file using Command Prompt.

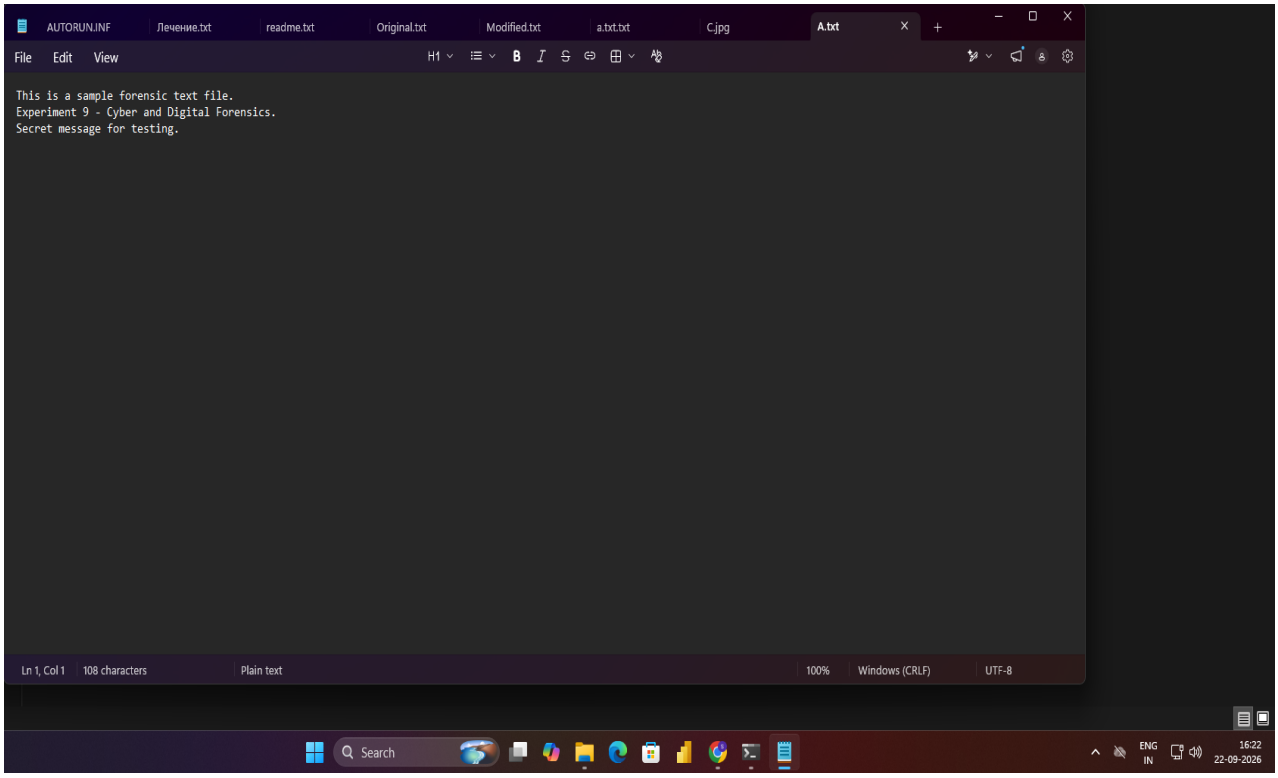
PROCEDURE

1. Copy the text file (A.txt) and image file (B.jpg) to the Desktop.
2. Open the Command Prompt by pressing **Win + R**, typing cmd, and pressing Enter.
3. Navigate to the Desktop using the command cd Desktop.
4. Use the command copy /b B.jpg + A.txt C.jpg to combine the image and text file.
5. Verify that the output file **C.jpg** has been created.
6. Open the output image file using Notepad.
7. Scroll to the end of the file to locate the hidden text content.
8. For audio files, place the audio file and the file to be hidden in the same folder.

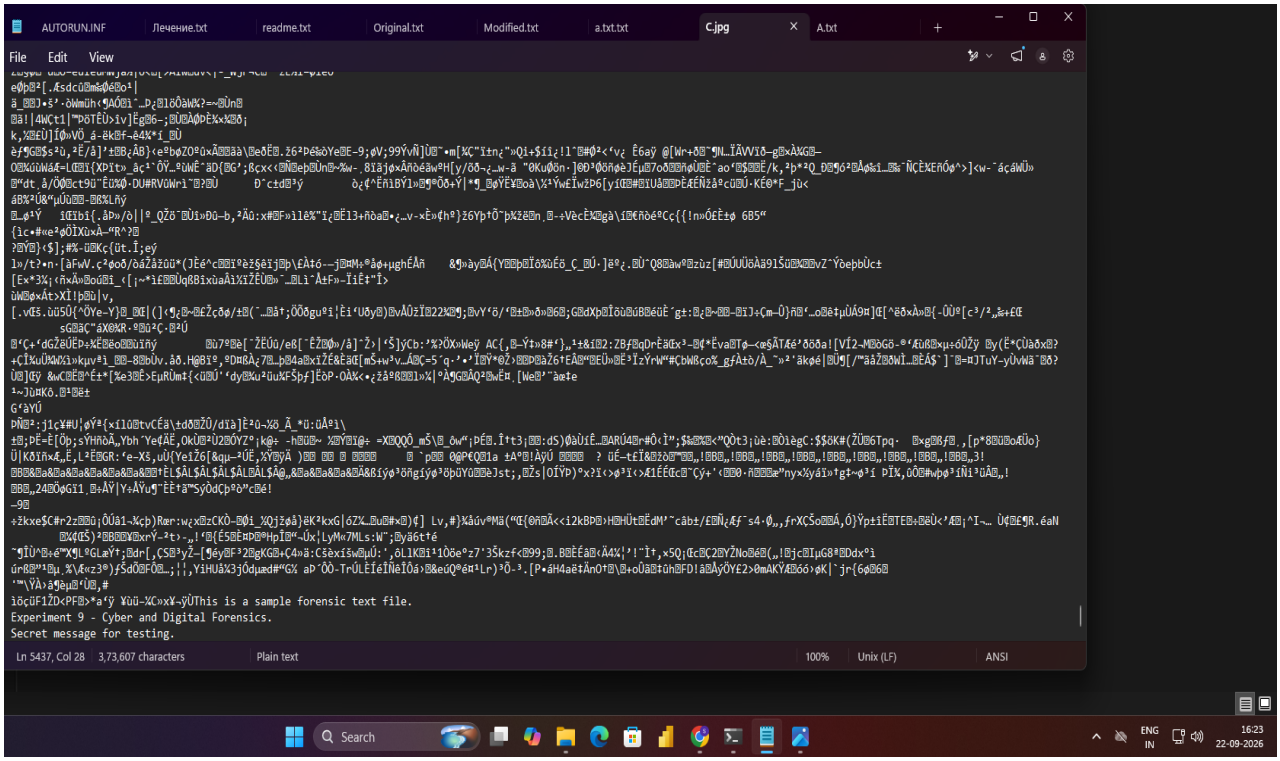
9. Use the binary copy command to combine the files and examine the output using a suitable archive utility.

OBSERVATION




AUTORUN.INF
Лечение.txt
readme.txt
Original.txt
Modified.txt
a.txt.txt
C.jpg
A.txt
File
Edit
View
This is a sample forensic text file.
Experiment 9 - Cyber and Digital Forensics.
Secret message for testing.
Ln 1, Col 1
108 characters
Plain text
100%
Windows (CRLF)
UTF-8
Search
ENG
IN
16:22
22-09-2026


Photos - FileInfo.com Example.jpg
Fullscreen
JPG file open in Microsoft Photos 2021. © FileInfo.com



RESULT

Thus, the steps for hiding and extracting text data behind image or audio files using Command Prompt were studied successfully.

NAME: SREE DHARSHINI M

ROLL NO: 241901110

EXPERIMENT 10

EXTRACT EXCHANGEABLE IMAGE FILE FORMAT (EXIF) DATA FROM IMAGE FILES USING EXIFREADER SOFTWARE

AIM

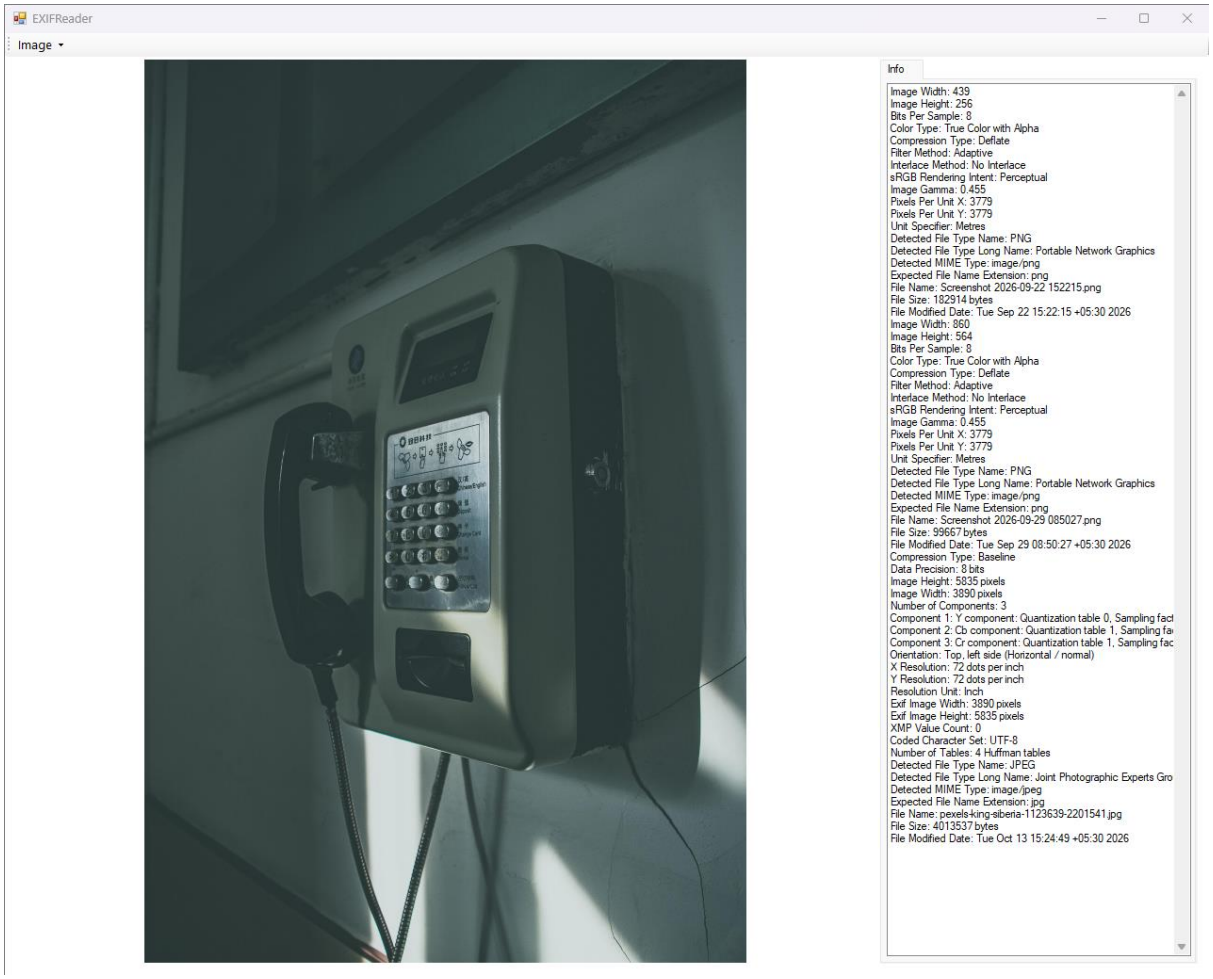
To extract and examine Exchangeable Image File Format (EXIF) data from image files using Exifreader software.

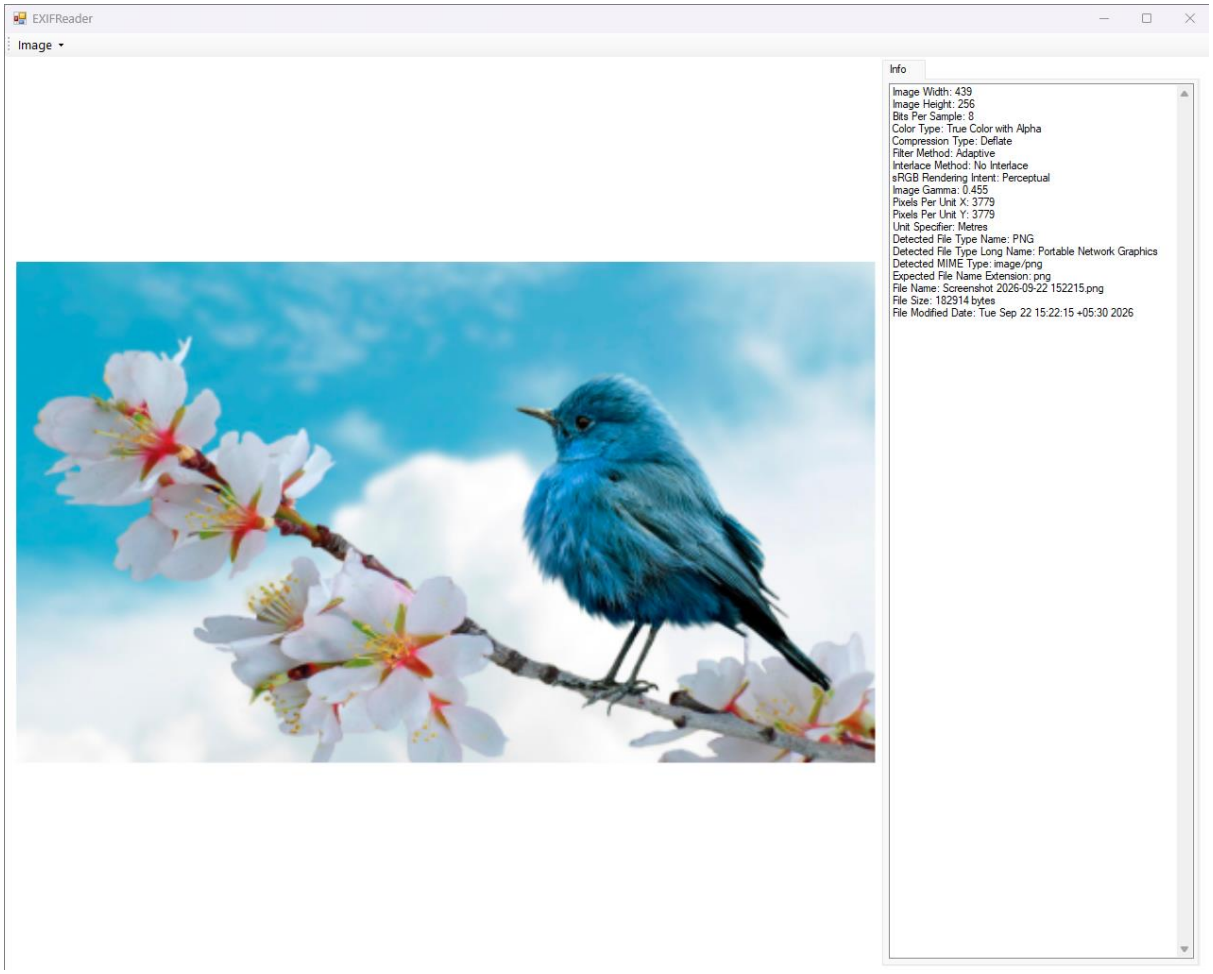
PROCEDURE

1. Select an image file from the computer for EXIF data extraction.
2. Download and install Exifreader software, if it is not already installed.
3. Open the Exifreader application.
4. Load or select the image file to be examined.
5. Allow the software to read and display the image's EXIF metadata.
6. Observe the camera make and model details, if available.
7. Examine image properties such as date and time, image dimensions, and file format.
8. Check other available metadata fields, such as exposure settings and orientation.

9. Record the displayed EXIF information and analyze the metadata obtained from the image.

OBSERVATION





RESULT

EXIF data was extracted and examined from an image file using Exifreader software.